CUA-Suite: 55 Hours of Expert Video for Teaching AI to Use Your Desktop

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ServiceNow released CUA-Suite, the largest open dataset for training AI agents that operate desktop computers. The core component, VideoCUA, contains roughly 10,000 human-demonstrated tasks across 87 professional applications (VS Code, Blender, LibreOffice, GNUCash, and others) recorded as continuous 30fps screen video totaling 55 hours and 6 million

frames. Each recording includes millisecond-precision cursor traces, keystroke logs, and multi-layered reasoning annotations averaging 497 words per step. Two companion resources round out the suite: GroundCUA, a grounding dataset with 56,000 annotated screenshots and 3.6 million UI element bounding boxes, and UI-Vision, a 450-task benchmark for evaluating how well models locate elements and predict actions. When tested on professional desktop software, current AI models fail around 60% of tasks.

The distinction between continuous video and sparse screenshots matters for anyone building desktop automation tools. Screenshots capture isolated moments. Video captures the full motion: how someone scans a menu, drags a slider, pauses to read feedback, then corrects course. That temporal signal is what lets an agent learn the dynamics of real interaction, not just "click here." At 2.5 times the size of the previous largest open dataset (ScaleCUA), CUA-Suite gives researchers the raw material to train agents on professional workflows instead of toy websites.

The 60% failure rate is the number to focus on. Desktop agents demo well on simple tasks but break on complex, multi-step professional workflows. CUA-Suite makes that gap measurable and, with its open data, closable. Princeton's OpenClaw-RL tackled unified agent training across modalities; CUA-Suite provides the desktop-specific data those training pipelines need.

Sources:

- CUA-Suite arXiv Paper
- CUA-Suite Project Page
- VideoCUA Dataset on HuggingFace
- HuggingFace Daily Papers: CUA-Suite

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